

FEATURE ENGINEERING DOCUMENTATION

1. Introduction

Feature engineering was conducted during week 3 to create additional variables that could provide more useful representations of the information contained in the original loan prediction dataset. The engineered features were developed from existing variables and evaluated based on their analytical meaning, distribution, relationship with other variables and potential usefulness for predictive modelling.

The main engineered features considered were:

- TotalIncome
- IncomeBand
- DebtToIncomeRatio
- FamilySizeGroup
- CreditRiskCategory

2. TotalIncome

Definition

TotalIncome was created by combining the applicant's income and coapplicant's income. This feature represents the combined income associated with a loan application.

Purpose

Using a combined income measure provides a broader representation of the financial resources associated with an application than considering ApplicantIncome and CoapplicantIncome separately.

Analytical findings

TotalIncome showed a strong positive correlation with ApplicantIncome.

The correlation between the two variables was:

$$r = 0.8930$$

This strong relationship is expected because TotalIncome incorporates ApplicantIncome directly. TotalIncome also showed a positive relationship with LoanAmount, with a correlation of approximately:

$$r = 0.6246$$

This suggests that applications with higher combined income tended to be associated with larger requested loan amounts.

Decision

TotalIncome was retained as an engineered numerical feature because it provides a meaningful representation of combined applicant income.

3. IncomeBand

Definition

IncomeBand was created by grouping TotalIncome into four categories:

- Low
- Lower-Middle
- Upper-Middle
- High

Purpose

The purpose of IncomeBand was to transform the continuous income measure into interpretable income groups that could be used for grouped comparisons and categorical analysis.

Distribution

The resulting distribution was:

Income Band	Count
Low	155
Lower-Middle	152
Upper-Middle	153
High	154

The groups were relatively balanced in size.

Statistical evaluation

The chi-square analysis between IncomeBand and Loan_Status produced a p-value of:

p = 0.889746

Since this value is greater than 0.05, there was insufficient statistical evidence of an association between IncomeBand and Loan_Status at the 5% significance level.

Decision

IncomeBand was retained in the final feature set because it provides an interpretable categorical representation of income and may still contribute useful information when combined with other predictors during machine-learning modelling.

4. DebtToIncomeRatio

Definition

DebtToIncomeRatio was engineered to represent the relationship between the requested loan amount and income. The feature provides an additional measure of the relative loan burden associated with an application.

Purpose

The purpose of this feature was to provide a financial indicator that considers loan amount relative to income rather than evaluating these variables independently.

Descriptive statistics

The resulting distribution was:

Statistic	Value
Count	592
Mean	23.755686
Standard deviation	8.295193
Minimum	2.522916
25th percentile	19.313443
Median	24.139119
75th percentile	28.148624
Maximum	82.711645

The feature contained some missing observations before the final refinement process.

Decision

DebtToIncomeRatio was retained as a numerical predictor because it provides an additional representation of the relationship between loan requirements and applicant income.

5. FamilySizeGroup

Definition

FamilySizeGroup was created to provide a group representation of family size based on the available dependent and family information.

The resulting categories were:

- Small Family
- Medium Family
- Large Family
- Unknown

Distribution

The feature had the following distribution:

Family Size Group	Count
Small Family	345
Medium Family	203
Large Family	51
Unknown	15

Statistical evaluation

The chi-square analysis between FamilySizeGroup and Loan_Status produced:

$p = 0.784945$

Since the p-value is greater than 0.05, there was insufficient evidence of a statistically significant association between FamilySizeGroup and Loan_Status.

Decision

FamilySizeGroup was retained as an engineered categorical feature because it provides an interpretable representation of family structure and can be evaluated alongside the other predictors during machine-learning modelling.

6. CreditRiskCategory

Definition

CreditRiskCategory was created from the applicant's Credit_History information to provide a categorical representation of credit risk.

Analytical findings

CreditRiskCategory showed a strong relationship with Loan_Status.

The chi-square test produced:

- $\chi^2 = 180.0665$
- $p < 0.001$

This indicates a strong statistical association between the feature and loan approval status.

Feature evaluation

Although CreditRiskCategory showed strong statistical evidence of association with Loan_Status, it was identified as redundant because it was derived directly from Credit_History. Both variables therefore represented essentially the same underlying information. Including both features could introduce unnecessary duplication into the modelling dataset.

Decision

CreditRiskCategory was removed from the final modelling dataset. Credit_History was retained as the original predictor because it contains the underlying information from which CreditRiskCategory was derived.

7. Summary of feature engineering decisions

Feature	Type	Decision	Reason
TotalIncome	Numerical	Retained	Provides combined income information
IncomeBand	Categorical	Retained	Provides interpretable income groups
DebtToIncomeRatio	Numerical	Retained	Represents loan burden relative to income
FamilySizeGroup	Categorical	Retained	Provides grouped family-size information

CreditRiskCategory	Categorical	Removed	Redundant with Credit_History
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8. Overall outcome

Feature engineering expanded the dataset beyond the original variables and provided additional representations of applicant financial circumstances and family characteristics. The engineered variables were not retained automatically. Each feature was considered in relation to its analytical usefulness, statistical findings, interpretability and redundancy. The most important refinement involved removing CreditRiskCategory because it duplicated information already contained in Credit_History. The remaining engineered features were incorporated into the final modelling dataset and subsequently encoded and scaled where applicable. The final modelling dataset contains 614 observations and 23 variables, consisting of 22 predictor variables and one target variable. Final validation confirmed that the refined dataset contains zero missing values and is ready for the machine-learning stage in week 4.